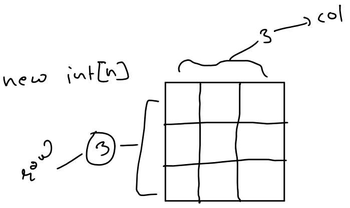
Lec31

01 April 2026 10:41 AM

2D Array

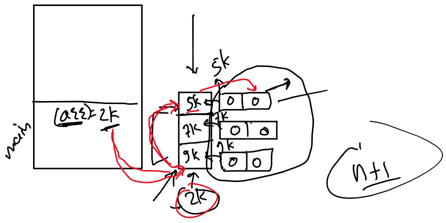
int arr[] = new int[n]



3 × 3

int $\underline{arr}$[][] = new int [3] [2]

row col



Total number of 1D Array = n+1

Sout(arr) ⟶ 2k

Sout(arr[0]) ⟶ 5k

Sout(arr[0][1]) ⟶ 0

[[I@543e710e
[I@57f23557
0

int row = sc.nextInt() ⟶ 3

int col = sc.nextInt() ⟶ 3

int arr[][] = new int[row][col]

	0	1	2
0	1 (0-0)	2 (0-1)	3 (0-2)
1	4 (1-0)	5 (1-1)	6 (1-2)
2	7 (2-0)	8 (2-1)	9 (2-2)

0 ⟶ arr[0][0] = 1
arr[0][1] = 2
arr[0][2] = 3

1 ⟶ arr[1][0] = 4
arr[1][1] = 5
arr[1][2] = 6

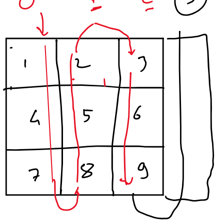
2 ⟶ arr[2][0] = 7
arr[2][1] = 8
arr[2][2] = 9

```
for(i=0; i<row; i++){
    for(j=0; j<col; j++){
        arr[i][j] = sc.nextInt()
    }
}
```

row = arr.length

col = arr[0].length

0 1 2 3 %2 = = 0



1 4 7 8 5 2 3 6 9

[0-0] [1-0] [2-0]

[2-1] [1-1] [0-1]

[0-2] [1-2] [2-2]

```
for(col = 0; col<arr[0].length; col++){
    if(col%2==0){
        for(row=0; row<arr.length; row++){
            // arr[row][col])
```

```
for(col - - ) -
    if(col%2==0){
        for(row=0; row<arr.length; row++){
            sout(arr[row][col])
        }
    }else{
        for(row=arr.length-1; row>=0; row--){
            sout(arr[row][col]
        }
    }
}
```

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

1	5	9	13
2	6	10	14
3	7	11	15
4	8	12	16

	0	1	2	3
0	1	5	9	13
1	2	6	10	15
2	3	7	11	15
3	4	8	12	16

0 → (0-1) ⟷ (1-0), (0-2) ⟷ (2-0), (0-3) ⟷ (3-0)

1 → (1-2) ⟷ (2-1), (1-3) ⟷ (3-1)

2 → (2-3) ⟷ (3-2),

3 → (3-4)

1 5 9 13
2 6 10 14
3 7 11 15
4 8 12 16